ExamGOAL
PYQ PDF

Free · With answer key

NEET · BIOLOGY · CHAPTER-WISE PYQS

Photosynthesis in Higher Plants

NEET Biology · Previous Year Questions with Answer Key

125	45	1988 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

MCQ (Single Correct Answer)	125 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

MCQ (Single Correct Answer) 125 questions

Q1. RE-NEET 2026

How many turns of Calvin cycle are required for the formation of three molecules of glucose?

(A) 18 (B) 6 (C) 3 (D) 1

Open solution

Q2. RE-NEET 2026

Photorespiration reaction catalyzed by RuBisCo is shown below:

$\mathrm{RuBP} + \mathrm{O_2} \rightarrow$ 3-Phosphoglycerate +X . Identify " X " from the given options:

(A) Malate (B) Phosphoenolpyruvate

(C) 2-Phosphoglycolate (D) Oxaloacetate

Open solution

Q3. RE-NEET 2026

Which pigment has absorption peak at 700 nm in the photosynthetic reaction centre PS I (P700)?

(A) Carotenoids (B) Chlorophyll b (C) Chlorophyll a (D) Xanthophylls

Open solution

Q4. RE-NEET 2026

Which of the following statements regarding photorespiration are correct?

(a) Do not occur in C 3 plants

(b) $\mathrm{CO_2}$ is consumed and $\mathrm{O_2}$ is generated

(c) Phosphoglycolate is formed

(d) No synthesis of ATP and NADPH

Choose the correct answer from the options given below:

(A) (a) and (b) only (B) (a) and (d) only (C) (c) and (d) only (D) (b) and (d) only

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q5. RE-NEET 2026

Select the correct sequence of experiments that led to a gradual understanding of photosynthesis in green plants.

(A) Production of glucose → role of air → release of oxygen → absorption spectra of chlorophyll a and b

(B) Absorption spectra of chlorophyll a and $\mathrm{b} \rightarrow$ production of glucose → release of oxygen $\rightarrow$ role of air

(C) Role of air → release of oxygen → production of glucose → absorption spectra of chlorophyll a and b

(D) Release of oxygen → production of glucose → absorption spectra of chlorophyll a and $b \rightarrow$ role of air

Open solution

Q6. NEET 2026

Find the incorrect statement(s) about photosynthesis from the following:

A. The water splitting complex is associated with PSI.

B. C_4 plants use the C_3 pathway of CO_2 fixation as the main biosynthetic pathway.

C. In C_4 plants, photorespiration does not occur.

D. C_3 plants exhibit 'Kranz' anatomy.

E. ATP synthesis in chloroplast occurs through chemiosmosis.

Choose the answer from the options given below:

(A) B and C only (B) B only (C) B and E only (D) A and D only

Open solution

Q7. NEET 2026

The enzyme required for carboxylation in the Calvin cycle is

(A) Hexokinase (B) PEP carboxylase

(C) RuBP carboxylase - oxygenase (D) Carboxypeptidase

Open solution

Q8. NEET 2026

How many ATP and NADPH molecules are required to make one molecule of glucose through the Calvin pathway?

(A) 18 ATP and 12 NADPH (B) 12 ATP and 18 NADPH

(C) 24 ATP and 18 NADPH (D) 6 ATP and 12 NADPH

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q9. NEET 2025

Which of the following statements about RuBisCO is true?

(A) It is an enzyme involved in the photolysis of water

(B) It catalyzes the carboxylation of RuBP

(C) It is active only in the dark

(D) It has higher affinity for oxygen than carbon dioxide

Open solution

Q10. NEET 2025

Match List-I with List-II

	List - I		List - II
A	Chlorophyll a	I	Yellow-green
B	Chlorophyll b	II	Yellow
C	Xanthophylls	III	Blue-green
D	Carotenoids	IV	Yellow to Yellow-orange

Choose the option with all correct matches.

(A) A-I, B-II, C-IV, D-III

(B) A-I, B-IV, C-III, D-II

(C) A-III, B-IV, C-II, D-I

(D) A-III, B-I, C-II, D-IV

Open solution

Q11. NEET 2024 (Re-Examination)

Which of the following are required for the light reaction of Photosynthesis?

A. CO_2

B. O_2

C. H_2O

D. Chlorophyll

E. Light

Choose the correct answer from the options given below:

(A) A, C, D and E only

(B) C, D and E only

(C) A and B only

(D) A, C and E only

Open solution

Q12. NEET 2024 (Re-Examination)

Which one of the following products diffuses out of the chloroplast during photosynthesis?

(A) ADP (B) NADPH (C) O_2 (D) ATP

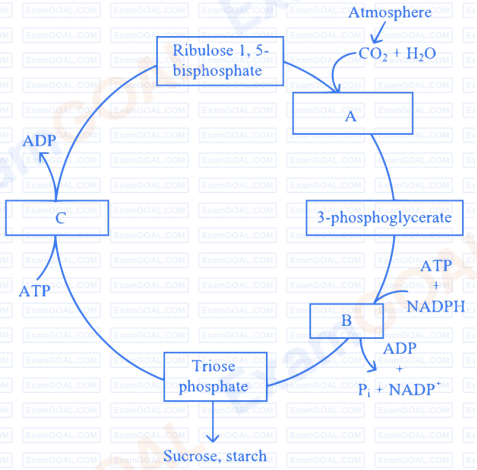
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q13. NEET 2024 (Re-Examination)

Observe the given figure. Identify the different stages labelled with alphabets by selecting the correct option.

Open solution



(A) A-Carboxylation, B-Regeneration, C-Reduction

(B) A-Reduction, B-Decarboxylation, C-Regeneration

(C) A-Carboxylation, B-Reduction, C-Regeneration

(D) A-Reduction, B-Carboxylation, C-Regeneration

Q14. NEET 2024

How many molecules of ATP and NADPH are required for every molecule of CO_2 fixed in the Calvin cycle?

Open solution

(A) 2 molecules of ATP and 3 molecules of NADPH

(B) 2 molecules of ATP and 2 molecules of NADPH

(C) 3 molecules of ATP and 3 molecules of NADPH

(D) 3 molecules of ATP and 2 molecules of NADPH

Q15. NEET 2024

Which of the following are required for the dark reaction of photosynthesis?

Open solution

A. Light

B. Chlorophyll

C. CO_2

D. ATP

E. NADPH

Choose the correct answer from the options given below:

(A) A, B and C only (B) B, C and D only (C) C, D and E only (D) D and E only

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q16. NEET 2024

Open solution

Given below are two statements:

Statement I: In C_3 plants, some O_2 binds to $RuBisCO$, hence CO_2 fixation is decreased.

Statement II: In C_4 plants, mesophyll cells show very little photorespiration while bundle sheath cells do not show photorespiration.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q17. NEET 2023 Manipur

Open solution

Match List - I with List - II.

	List - I		List - II
(A)	Chlorophyll a	(I)	Yellow to yellow orange
(B)	Chlorophyll b	(II)	Yellow green
(C)	Xanthophyll	(III)	Blue green
(D)	Carotenoid	(IV)	Yellow

Choose the correct answer from the options given below :

(A) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(B) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(C) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(D) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

Q18. NEET 2023 Manipur

Open solution

Given below are two statements :

Statement I : RuBisCO is the most abundant enzyme in the world.

Statement II : Photorespiration does not occur in C_4 plants.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but Statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are correct

(D) Both Statement I and Statement II are incorrect

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q19. NEET 2023 Manipur

Which out of the following statements is incorrect?

(A) Grana lamellae have both PS I and PS II

(B) Cyclic photophosphorylation involves both PS I and PS II

(C) Both ATP and NADPH + H^+ are synthesised during non-cyclic photophosphorylation.

(D) Stroma lamellae lack PS II and NADP reductase

Open solution

Q20. NEET 2023

How many ATP and $NADPH_2$ are required for the synthesis of one molecule of Glucose during Calvin cycle?

(A) 18 ATP and 12 $NADPH_2$

(B) 12 ATP and 16 $NADPH_2$

(C) 18 ATP and 16 $NADPH_2$

(D) 12 ATP and 12 $NADPH_2$

Open solution

Q21. NEET 2023

The reaction centre in PS II has an absorption maxima at

(A) 700 nm (B) 660 nm (C) 780 nm (D) 680 nm

Open solution

Q22. NEET 2023

Which of the following combinations is required for chemiosmosis?

(A) Membrane, proton pump, proton gradient, NADP synthase

(B) Proton pump, electron gradient, ATP synthase

(C) Proton pump, electron gradient, NADP synthase

(D) Membrane, proton pump, proton gradient, ATP synthase

Open solution

Q23. NEET 2022 Phase 2

When one CO_2 molecule is fixed as one molecule of triose phosphate, which of the following photochemically made, high energy chemical intermediates are used in the reduction phase?

(A) 2 ATP + 2 NADPH (B) 1 ATP + 1 NADPH (C) 1 ATP + 2 NADPH (D) 2 ATP + 1 NADPH

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q24. NEET 2022 Phase 2

Identify the correct statements regarding chemiosmotic hypothesis:

(a) Splitting of the water molecule takes place on the inner side of the membrane.

(b) Protons accumulate within the lumen of the thylakoids.

(c) Primary acceptor of electron transfers the electrons to an electron carrier.

(d) NADP reductase enzyme is located on the stroma side of the membrane.

(e) Protons increase in number in stroma.

Choose the correct answer from the options given below:

(A) (b), (c) and (e) **(B)** (a), (b) and (e) **(C)** (a), (b) and (d) **(D)** (b), (c) and (d)

Open solution

Q25. NEET 2022 Phase 1

Given below are two statements:

Statement I : The primary CO_2 acceptor in C_4 plants is phosphoenolpyruvate and is found in the mesophyll cells.

Statement II : Mesophyll cells of C_4 plants lack RuBisCo enzyme.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Open solution

Q26. NEET 2022 Phase 1

Which one of the following is not true regarding the release of energy during ATP synthesis through chemiosmosis? It involves:

(A) Breakdown of proton gradient

(B) Breakdown of electron gradient

(C) Movement of protons across the membrane to the stroma

(D) Reduction of NADP to $NADPH_2$ on the stroma side of the membrane

Open solution

Q27. NEET 2021

The first stable product of CO_2 fixation in Sorghum is

(A) Phosphoglyceric acid **(B)** Pyruvic acid

(C) Oxaloacetic acid **(D)** Succinic acid

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q28. NEET 2021

Which of the following statements is incorrect?

(A) Cyclic photophosphorylation involves both PS I and PS II

(B) Both ATP and NADPH + H^+ are synthesized during non-cyclic photophosphorylation.

(C) Stoma lamellae have PS I only and lack NADP reductase

(D) Grana lamellae have both PS I and PS II

Open solution

Q29. NEET 2020 Phase 2

Which of the following statements is **incorrect** ?

(A) RuBisCO action requires ATP and NADPH

(B) RuBisCO is a bifunctional enzyme

(C) In C_4 plants, the site of RuBisCO activity is mesophyll cell

(D) The substrate molecule for RuBisCO activity is a 5-carbon compound

Open solution

Q30. NEET 2020 Phase 2

During non-cyclic photophosphorylation, when electrons are lost from the reaction centre at PS II, what is the source which replaces these electrons ?

(A) Light (B) Oxygen (C) Water (D) Carbon dioxide

Open solution

Q31. NEET 2020 Phase 1

In light reaction, plastoquinone facilitates the transfer of electrons from

(A) PS-I to ATP synthase

(B) $Cytb_6f$ complex to PS-I

(C) PS-I to $NADP^+$

(D) PS-II to $Cytb_6f$ complex

Open solution

Q32. NEET 2020 Phase 1

The oxygenation activity of RuBisCo enzyme in photorespiration leads to the formation of

(A) 1 molecule of 3-C compound and 1 molecule of 2-C compound

(B) 1 molecule of 3-C compound

(C) 1 molecule of 6-C compound

(D) 2 molecules of 3-C compound

Open solution

Q33. NEET 2019 (Odisha)

In Hatch and Slack pathway, the primary CO_2 acceptor is

(A) Rubisco

(B) Oxaloacetic acid

(C) Phosphoglyceric acid

(D) Phosphoenol pyruvate

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q34. NEET 2019 (Odisha)

One scientist cultured *Cladophora* in a suspension of *Azotobacter* and illuminated the culture by splitting light through a prism. He observed that bacteria accumulated mainly in the region of :

(A) Blue and red light
(B) Violet and green light
(C) Indigo and green light
(D) Orange and yellow light

Open solution

Q35. NEET 2018

Which of the following is not a product of light reaction of photosynthesis?

(A) Oxygen
(B) NADH
(C) ATP
(D) NADPH

Open solution

Q36. NEET 2017

With reference to factors affecting the rate of photosynthesis, which of the following statements is not correct?

(A) Increasing atmospheric CO_2 concentration up to 0.05% can enhance CO_2 fixation rate.
(B) C_3 plants respond to higher temperature with enhanced photosynthesis while C_4 plants have much lower temperature optimum.
(C) Tomato is a greenhouse crop which can be grown in CO_2 -enriched atmosphere for higher yield.
(D) Light saturation for CO_2 fixation occurs at 10% of full sunlight.

Open solution

Q37. NEET 2017

Phosphoenol pyruvate (PEP) is the primary CO_2 acceptor in

(A) C_4 plants
(B) C_2 plants
(C) C_3 and C_4 plants
(D) C_3 plants

Open solution

Q38. NEET 2016 Phase 2

The process which makes major difference between C_3 and C_4 plants is

(A) glycolysis
(B) Calvin cycle
(C) photorespiration
(D) respiration.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q39. NEET 2016 Phase 1

Water vapour comes out from the plant leaf through the stomatal opening. Through the same stomatal opening carbon dioxide diffuses into the plant during photosynthesis. Reason out the above statements using one of following options.

(A) The above processes happen only during night time.

(B) One process occurs during day time and the other at night.

(C) Both processes cannot happen simultaneously.

(D) Both processrs can happen together because the diffusion coefficient of water and CO_2 is different.

Open solution

Q40. NEET 2016 Phase 1

In a chloroplast the highest number of protons are found in

(A) intermembrane space

(B) antennae complex

(C) stroma

(D) lumen of thylakoids.

Open solution

Q41. NEET 2016 Phase 1

Emerson's enhancement effect and Red drop have been instrumental in the discovery of

(A) photophosphorylation and cyclic electron transport

(B) oxidative phosphorylation

(C) photophosphorylation and non-cyclic electron transport

(D) two photosystems operating simultaneously.

Open solution

Q42. NEET 2016 Phase 1

A plant in your garden avoids photorespiratory losses, has improved water use efficiency, shows high rates of photosynthesis at high temperatures and has improved efficiency of nitrogen utilisation. In which of the following physiological groups would you assign this plant?

(A) CAM

(B) Nitrogen fixer

(C) C_3

(D) C_4

Open solution

Q43. NEET 2016 Phase 1

Water soluble pigments found in plant cell vacuoles are

(A) carotenoids

(B) anthocyanins

(C) xanthophylls

(D) chlorophylls.

Open solution

Q44. AIPMT 2015

In photosynthesis, the light-independent reactions take place at

(A) photosystem II

(B) stromal matrix

(C) thylakoid lumen

(D) photosystem I.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q45. AIPMT 2015

Chromatophores take part in

(A) movement (B) respiration (C) photosynthesis (D) growth.

Open solution

Q46. AIPMT 2012 Mains

Read the following four statements (A – D). phosphorylation involve uphill transport of protons across the membrane.

(A) Both photophosphorylation and oxidative phosphorylation involve uphill transport of protons across the membrane.

(B) In dicot stems, a new cambium originates from cells of pericycle at the time of secondary growth.

(C) Stamens in flowers of *Gloriosa* and *Petunia* are polyandrous.

(D) Symbiotic nitrogen fixers occur in free-living state also in soil.

How many of the above statements are right?

(A) Two (B) Three (C) Four (D) One

Open solution

Q47. AIPMT 2012 Prelims

A process that makes important difference between C_3 and C_4 plants is

(A) transpiration (B) glycolysis

(C) photosynthesis (D) photorespiration.

Open solution

Q48. AIPMT 2012 Prelims

The correct sequence of cell organelles during photorespiration is

(A) chloroplast, Golgi-bodies, mitochondria

(B) chloroplast, rough endoplasmic reticulum, dictyosomes

(C) chloroplast, peroxisome, mitochondria

(D) Chloroplast, vacuole, peroxisome.

Open solution

Q49. AIPMT 2011 Mains

In Kranz anatomy, the bundle sheath cells have

(A) thin walls, many intercellular spaces and no chloroplasts

(B) thick walls, no intercellular spaces and large number of chloroplasts

(C) thin walls, no intercellular spaces and several chloroplasts

(D) thick walls, many intercellular spaces and few chloroplasts.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q50. AIPMT 2011 Mains

Which one of the following is essential for photolysis of water?

(A) Copper (B) Manganese (C) Boron (D) Zinc

Open solution

Q51. AIPMT 2011 Prelims

CAM helps the plants in

(A) conserving water (B) secondary growth

(C) disease resistance (D) reproduction.

Open solution

Q52. AIPMT 2010 Mains

Read the following four statements, (i), (ii), (iii) and (iv) and select the right option having both correct statements.

Statements :

(i) Z scheme of light reaction takes place in presence of PSI only.

(ii) Only PSI is functional in cyclic photophosphorylation.

(iii) Cyclic photophosphorylation results into synthesis of ATP and $NADPH_2$.

(iv) Stroma lamellae lack PSII as well as NADP.

(A) (ii) and (iv) (B) (i) and (ii) (C) (ii) and (iii) (D) (iii) and (iv).

Open solution

Q53. AIPMT 2010 Mains

Kranz anatomy is one of the characteristics of the leaves of

(A) potato (B) wheat (C) sugarcane (D) mustard.

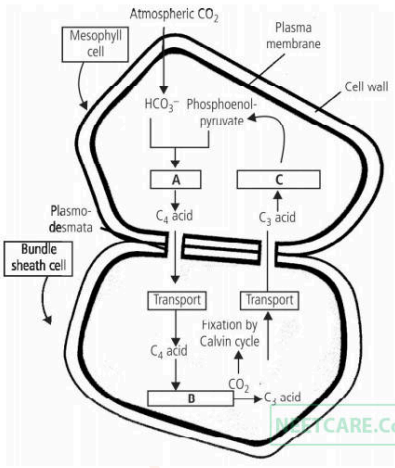
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q54. AIPMT 2010 Mains

Open solution

Study the pathway given below:



In which of the following options correct words for all the three blanks A, B and C are indicated?

(A) A → Fixation, B → Transamination, C → Regeneration

(B) A → Decarboxylation, B → Reduction, C → Regeneration

(C) A → Carboxylation, B → Decarboxylation, C → Reduction

(D) A → Fixation, B → Decarboxylation, C → Regeneration

Q55. AIPMT 2010 Prelims

Open solution

PGA as the first CO_2 fixation product was discovered in photosynthesis of

(A) bryophyte (B) gymnosperm (C) angiosperm (D) alga.

Q56. AIPMT 2010 Prelims

Open solution

C_4 plants are more efficient in photosynthesis than C_3 plants due to

(A) higher leaf area

(B) presence of larger number of chloroplasts in the leaf cells

(C) presence of thin cuticle

(D) lower rate of photorespiration.

Q57. AIPMT 2009

Open solution

Cyclic photophosphorylation results in the formation of

(A) ATP and NADPH (B) ATP, NADPH and O_2

(C) ATP (D) NADPH.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q58. AIPMT 2009

Stroma in the chloroplasts of higher plants contains

(A) light-dependent reaction enzymes (B) ribosomes

(C) chlorophyll (D) light-independent reaction enzymes.

Open solution

Q59. AIPMT 2008

Electrons from excited chlorophyll molecule of photosystem II are accepted first by

(A) quinone (B) ferredoxin (C) cytochrome-*b* (D) cytochrome-*f*.

Open solution

Q60. AIPMT 2008

The C_4 plants are photosynthetically more efficient than C_3 plants because

(A) the CO_2 efflux is not prevented

(B) they have more chloroplasts

(C) the CO_2 compensation point is more

(D) CO_2 generated during photorespiration is trapped and recycled through PEP carboxylase.

Open solution

Q61. AIPMT 2008

In leaves of C_4 plants malic acid synthesis during CO_2 fixation occurs in

(A) bundle sheath (B) guard cells (C) epidermal cells (D) mesophyll cells.

Open solution

Q62. AIPMT 2007

In the leaves of C_4 plants, malic acid formation during CO_2 fixation occurs in the cells of

(A) bundle sheath (B) phloem (C) epidermis (D) mesophyll.

Open solution

Q63. AIPMT 2007

The first acceptor of electrons from an excited chlorophyll molecule of photosystem II is

(A) iron-sulphur protein (B) ferredoxin

(C) quinone (D) cytochrome.

Open solution

Q64. AIPMT 2006

During photorespiration, the oxygen consuming reaction(s) occur in

(A) stroma of chloroplasts (B) stroma of chloroplasts and mitochondria

(C) stroma of chloroplasts and peroxisomes (D) grana of chloroplasts and peroxisomes.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q65. AIPMT 2006

In photosystem I, the first electron acceptor is

(A) an iron-sulphur protein
(B) ferredoxin
(C) cytochrome
(D) plastocyanin.

Open solution

Q66. AIPMT 2005

As compared to a C_3-plant, how many additional molecules of ATP are needed for net production of one molecule of hexose sugar by C_4-plants?

(A) Two
(B) Six
(C) Twelve
(D) Zero

Open solution

Q67. AIPMT 2005

Photosynthetic Active Radiation (PAR) has the following range of wavelengths.

(A) 340-450 nm
(B) 400-700 nm
(C) 500-600 nm
(D) 450-950 nm

Open solution

Q68. AIPMT 2005

Photosynthesis in C_4 plants is relatively less limited by atmospheric CO_2 levels because

(A) effective pumping of CO_2 into bundle sheath cells
(B) RuBisCo in C_4 plants has higher affinity for CO_2
(C) four carbon acids are the primary initial CO_2 fixation products
(D) the primary fixation of CO_2 is mediated *via* PEP carboxylase.

Open solution

Q69. AIPMT 2004

In C_3 plants, the first stable product of photosynthesis during the dark reaction is

(A) malic acid
(B) oxaloacetic acid
(C) 3-phosphoglyceric acid
(D) phosphoglyceraldehyde.

Open solution

Q70. AIPMT 2004

Plants adapted to low light intensity have

(A) larger photosynthetic unit size than the sun plants
(B) higher rate of CO_2 fixation than the sun plants
(C) more extended root system
(D) leaves modified to spines.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q71. AIPMT 2004

Photosynthetically Active Radiation (PAR) represents the following range of wavelength

(A) 500-600 nm (B) 450-950 nm (C) 340-450 nm (D) 400-700 nm

Open solution

Q72. AIPMT 2003

Which fractions of the visible spectrum of solar radiations are primarily absorbed by carotenoids of the higher plants?

(A) Blue and green (B) Green and red (C) Red and violet (D) Violet and blue.

Open solution

Q73. AIPMT 2003

Which one of the following is wrong in relation to photorespiration?

(A) It occurs in chloroplast (B) It occurs in day time only

(C) It is a characteristic of C_4 plants (D) It is a characteristic of C_3 plants.

Open solution

Q74. AIPMT 2003

In sugarcane plant $^{14}CO_2$ is fixed to malic acid, in which the enzyme that fixes CO_2 is

(A) ribulose biphosphate carboxylase (B) phosphoenal pyruvic acid carboxylase

(C) ribulose phosphate kinase (D) fructose phosphate.

Open solution

Q75. AIPMT 2003

Stomata of CAM plants

(A) are always open

(B) open during the day and close at night

(C) open during the night and close during the day

(D) never open.

Open solution

Q76. AIPMT 2003

Which element is located at the centre of the porphyrin ring in chlorophyll?

(A) Calcium (B) Magnesium (C) potassium (D) Manganese

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q77. AIPMT 2003

Which one of the following concerns photophosphorylation?

(A) ADP + AMP $\xrightarrow{Light\ energy}$ ATP

(B) ADP + Inorganic PQ_4 $\xrightarrow{Light\ energy}$ ATP

(C) ADP + Inorganic PQ_4 $\longrightarrow$ ATP

(D) AMP + Inorganic PQ_4 $\xrightarrow{Light\ energy}$ ATP

Open solution

Q78. AIPMT 2002

Which of the following absorbs light energy for photosynthesis?

(A) Chlorophyll (B) Water molecule (C) O_2 (D) RuBP

Open solution

Q79. AIPMT 2002

In photosynthesis energy from light reaction to dark reaction is transferred in the form of

(A) ADP (B) ATP (C) RuDP (D) chlorophyll.

Open solution

Q80. AIPMT 2002

Which pigment absorbs the red and far-red light?

(A) Cytochrome (B) Phytochrome (C) Carotenoids (D) Chlorophyll.

Open solution

Q81. AIPMT 2001

What is true for photolithotrophs?

(A) Obtain energy from radiations and hydrogen from organic compounds

(B) Obtain energy from radiations and hydrogen from inorganic compounds

(C) Obtain energy from organic compounds

(D) Obtain energy from inorganic compounds

Open solution

Q82. AIPMT 2001

Which pigment system is inactivated in red drop?

(A) PS-I and PS-II (B) PS-I

(C) PS-II (D) None of the above

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q83. AIPMT 2001

Which pair is wrong?

(A) C_3-maize

(B) C_4-kranz anatomy

(C) Calvin cycle-PGA

(D) Hatch and Slack cycle - OAA

Open solution

Q84. AIPMT 2000

Which is the first CO_2 acceptor enzyme in C_4 plants?

(A) RuDP carboxylase (B) Phosphoric acid (C) RuBisCO (D) PEP- carboxylase

Open solution

Q85. AIPMT 2000

For assimilation of one CO_2 molecule, the energy required in form of ATP and $NADPH_2$ are

(A) 2 ATP and 2 $NADPH_2$

(B) 5 ATP and 3 $NADPH_2$

(C) 3 ATP and 2 $NADPH_2$

(D) 18 ATP and 12 $NADPH_2$.

Open solution

Q86. AIPMT 2000

For the synthesis of one glucose molecule the Calvin cycle operates for

(A) 2 times (B) 4 times (C) 6 times (D) 8 times

Open solution

Q87. AIPMT 2000

The first step for initiation of photosynthesis will be

(A) photolysis of water

(B) excitement of chlorophyll molecules due to absorption of ligtht

(C) ATP formation

(D) glucose formation

Open solution

Q88. AIPMT 2000

Photochemical reactions in the chloroplast are directly involved in

(A) photolysis of water and phosphorylation of ADP to ATP

(B) formation of phosphoglyceric acid

(C) synthesis of glucose and starch

(D) fixation of carbon dioxide

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q89. AIPMT 1999

Carbon dioxide acceptor in C_3-plants is

(A) PGA (B) RuDP (C) PEP (D) none of these

Open solution

Q90. AIPMT 1999

The rate of photosynthesis is higher in

(A) red light (B) blue light (C) green light (D) very high light

Open solution

Q91. AIPMT 1997

Chlorophyll *a* molecule at its carbon atom 3 of the pyrrole ring II has one of the following

(A) carboxylic group (B) magnesium (C) aldehyde group (D) methyl group

Open solution

Q92. AIPMT 1997

The core metal of chlorophyll is

(A) Ni (B) Cu (C) Fe (D) Mg

Open solution

Q93. AIPMT 1997

$NADPH_2$ is generated through

(A) photosystem II (B) anaerobic respiration
(C) glycolysis (D) photosystem I

Open solution

Q94. AIPMT 1997

The mechanism of ATP formation both in chloroplast and mitochondria is explained by

(A) chemiosmotic theory (B) Munch's hypothesis (mass flow model)
(C) relay pump theory of Godlewski (D) Cholodny-Wont's model

Open solution

Q95. AIPMT 1996

'The law of limiting factors' was proposed by

(A) Leibig (B) Hatch and Slack (C) Blackman (D) Arnon

Open solution

Q96. AIPMT 1996

What will be the number of Calvin cycles to generate one molecule of hexose?

(A) 8 (B) 9 (C) 4 (D) 6

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q97. AIPMT 1996

Photorespiration is favoured by

(A) high temperature and low O_2
(B) high humidity and temperature
(C) high O_2 and low CO_2
(D) high CO_2 and low O_2

Open solution

Q98. AIPMT 1996

In C_4 plants, CO_2 combines with

(A) phosphoenol pyruvate
(B) phosphoglyceraldehyde
(C) phosphoglyceric acid
(D) ribulose diphosphate

Open solution

Q99. AIPMT 1996

In C_4 plants, CO_2 fixation is done by

(A) sclerenchyma
(B) chlorenchyma and hypodermis
(C) mesophyll cells
(D) guard cells

Open solution

Q100. AIPMT 1995

The primary acceptor, during CO_2 fixation in C_3 plants, is

(A) phosphoenolpyruvate (PEP)
(B) ribulose 1, 5-diphosphate (RuDP)
(C) phosphoglyceric acid (PGA)
(D) ribulose monophosphate (RMP)

Open solution

Q101. AIPMT 1995

The CO_2 fixation during C_4 pathway occurs in the chloroplast of

(A) guard cells
(B) bundle sheath cells
(C) mesophyll cells
(D) spongy parenchyma

Open solution

Q102. AIPMT 1994

Which of the following pigments acts as a reaction-centre during photosynthesis?

(A) Carotene (B) Phytochrome (C) P_{700} (D) Cytochrome

Open solution

Q103. AIPMT 1994

During light reaction of photosynthesis, which of the following phenomenon is observed during cyclic phosphorylation as well as non-cyclic phosphorylation?

(A) Release of O_2
(B) Formation of ATP
(C) Formation of NADPH
(D) Involvement of PS I and PS II pigment systems

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q104. AIPMT 1993

A photosynthesising plant is releasing ^{18}O more than the normal. The plant must have been supplied with

(A) O_3
(B) H_2O with ^{18}O
(C) CO_2 with ^{18}O
(D) $C_6H_{12}O_6$ with ^{18}O

Open solution

Q105. AIPMT 1993

Maximum solar energy is trapped by

(A) planting trees
(B) cultivating crops
(C) growing algae in tanks
(D) growing grasses

Open solution

Q106. AIPMT 1993

The carbon dioxide acceptor in Calvin cycle/C_3-plants is

(A) phosphoenol pyruvate (PEP)
(B) ribulose 1, 5-diphosphate (RuDP)
(C) phosphoglyceric acid (PGA)
(D) ribulose monophosphate (RMP)

Open solution

Q107. AIPMT 1993

Which one is a C_4-plant?

(A) Papaya (B) Pea (C) Potato (D) Maize/corn

Open solution

Q108. AIPMT 1992

Chlorophyll *a* occurs in

(A) all photosynthetic autotrophs
(B) in all higher plants
(C) all oxygen liberating autotrophs
(D) all plants except fungi

Open solution

Q109. AIPMT 1992

Photosystem II occurs in

(A) stroma
(B) cytochrome
(C) grana
(D) mitochondrial surface

Open solution

Q110. AIPMT 1992

The enzyme that catalyses carbon dioxide fixation in C_4 plants is

(A) RuBP carboxylase
(B) PEP carboxylase
(C) carbonic anhydrase
(D) carboxydismutase

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q111. AIPMT 1992

The first carbon dioxide acceptor in C_4-plants is

(A) phosphoenol-pyruvate
(B) ribulose 1, 5-diphosphate
(C) oxaloacetic acid
(D) phosphoglyceric acid

Open solution

Q112. AIPMT 1992

At a temperature above $35^{\circ}C$

(A) rate of photosynthesis will decline earlier than that of respiration
(B) rate of respiration will decline earlier than that of photosynthesis
(C) there is no fixed pattern
(D) both decline simultaneously

Open solution

Q113. AIPMT 1991

Ferredoxin is a constituent of

(A) PS I
(B) PS II
(C) Hill reaction
(D) P_{680}

Open solution

Q114. AIPMT 1991

During monsoon, the rice crop of eastern states of India shows lesser yield due to limiting factor of

(A) CO_2
(B) light
(C) temperature
(D) water

Open solution

Q115. AIPMT 1991

Which technique has helped in investigation of Calvin cycle?

(A) X-ray crystallography
(B) X-ray technique
(C) Radioactive isotope technique
(D) Intermittent light

Open solution

Q116. AIPMT 1991

Dark reactions of photosynthesis occur in

(A) granal thylakoid membranes
(B) stromal lamella membranes
(C) stroma outside photosynthetic lamellae
(D) periplastidial space

Open solution

Q117. AIPMT 1991

Photosynthetic pigments found in the chloroplasts occur in

(A) thylakoid membranes
(B) plastoglobules
(C) matrix
(D) chloroplast envelope

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q118. AIPMT 1990

Kranz anatomy is typical of

(A) C_4 plants (B) C_3 plants (C) C_2 plants (D) CAM plants

Open solution

Q119. AIPMT 1990

The first carbon dioxide acceptor in C_4-plants is

(A) phosphoenol-pyruvate (B) ribulose 1, 5-diphosphate
(C) oxaloacetic acid (D) phosphoglyceric acid

Open solution

Q120. AIPMT 1989

A very efficient converter of solar energy with net productivity of 204 kg/m^2 or more is the crop

(A) wheat (B) sugarcane (C) rice (D) bajra

Open solution

Q121. AIPMT 1989

In C_4 plants, Calvin cycle operates in

(A) stroma of bundle sheath chloroplasts (B) grana of bundle sheath chloroplasts
(C) grana of mesophyll chloroplasts (D) stroma of mesophyll chloroplasts

Open solution

Q122. AIPMT 1989

The substrate for photorespiration is

(A) phosphoglyceric acid (B) glycolate
(C) serine (D) glycine

Open solution

Q123. AIPMT 1989

The size of chlorophyll molecule is

(A) head 15 × 15 Å, tail 25 Å (B) head 20 × 20 Å, tail 25 Å
(C) head 15 × 15 Å, tail 20 Å (D) head 10 × 12 Å, tail 25 Å

Open solution

Q124. AIPMT 1988

$NADP^+$ is reduced to NADPH in

(A) PS I (B) PS II
(C) Calvin cycle (D) noncyclic photophosphorylation

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q125. AIPMT 1988

Carbon dioxide joins the photosynthetic pathway in

(A) PS I (B) PS II (C) light reaction (D) dark reaction

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

1. **A**	2. **C**	3. **C**	4. **C**	5. **C**
6. **D**	7. **C**	8. **A**	9. **B**	10. **D**
11. **B**	12. **C**	13. **C**	14. **D**	15. **C**
16. **C**	17. **A**	18. **C**	19. **B**	20. **A**
21. **D**	22. **D**	23. **A**	24. **C**	25. **A**
26. **B**	27. **C**	28. **A**	29. **C**	30. **C**
31. **D**	32. **A**	33. **D**	34. **A**	35. **B**
36. **B**	37. **A**	38. **C**	39. **D**	40. **D**
41. **D**	42. **D**	43. **B**	44. **B**	45. **C**
46. **A**	47. **D**	48. **C**	49. **B**	50. **B**
51. **A**	52. **A**	53. **C**	54. **D**	55. **D**
56. **D**	57. **C**	58. **D**	59. **A**	60. **B**
61. **D**	62. **D**	63. **C**	64. **C**	65. **A**
66. **C**	67. **B**	68. **D**	69. **C**	70. **A**
71. **D**	72. **D**	73. **C**	74. **B**	75. **C**
76. **B**	77. **B**	78. **A**	79. **B**	80. **B**
81. **B**	82. **C**	83. **A**	84. **D**	85. **C**
86. **C**	87. **B**	88. **A**	89. **B**	90. **A**
91. **D**	92. **D**	93. **D**	94. **A**	95. **C**
96. **D**	97. **C**	98. **A**	99. **C**	100. **B**
101. **C**	102. **C**	103. **B**	104. **B**	105. **C**
106. **B**	107. **D**	108. **C**	109. **C**	110. **B**
111. **A**	112. **A**	113. **A**	114. **B**	115. **C**
116. **C**	117. **A**	118. **A**	119. **A**	120. **B**
121. **A**	122. **B**	123. **C**	124. **D**	125. **D**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs